



Call Center

Analysis

Here is your task
It's omnipresent: telecom marketing. Better price here. Better service there. Best for small businesses here. Best for young urbanites there. But what do customers really want? Our client, a big telecom company needs to know. This email just arrived for you:

Request: Call Centre Trends

 Claire (PhoneNow)
To ● You

↩ Reply

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⋮

Hi Digital Accelerator,

May I introduce myself? I'm Claire, Call Centre Manager here at PhoneNow. My colleague suggested I reach out to you. Pleased to meet you.

I'm looking for transparency and insight into the data we have here at the Call Centre. For example: total number of calls answered and abandoned, speed of answer, length of calls, overall customer satisfaction, etc. What I'm after is an accurate overview of long-term trends in customer and agent behaviour.

I was told you are great at visualising data in such a way that important aspects become very clear. That's precisely what I need. Could you please prepare a dashboard on Call Centre trends that I can use as a basis for discussion with management? I'll provide you with the required data, of course.

Looking forward to hearing from you.

Best regards,
Claire

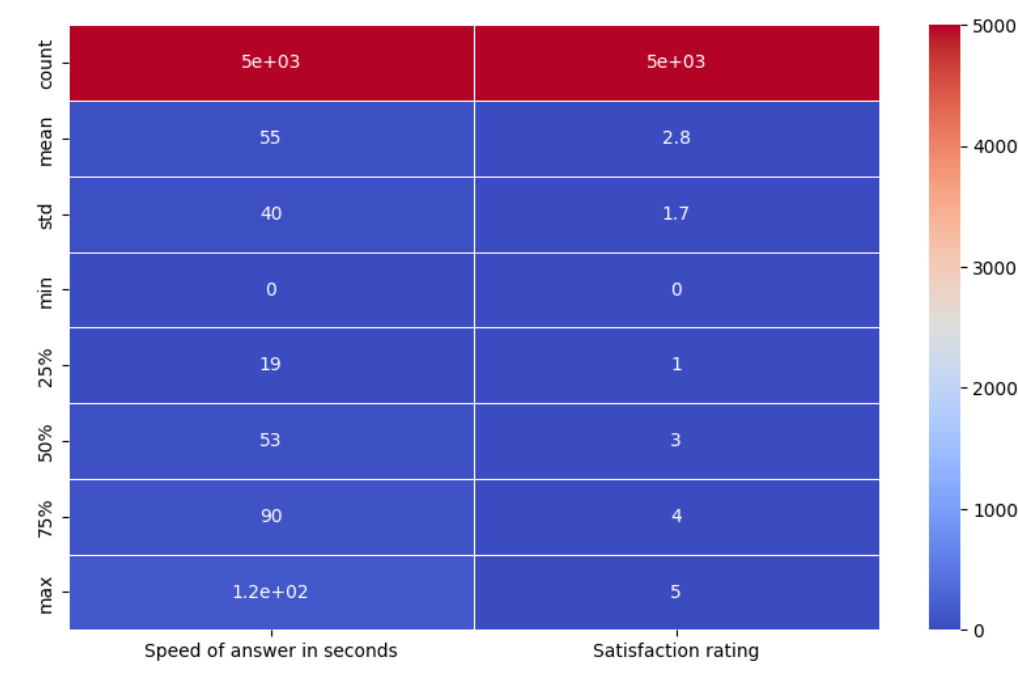
- Overall customer satisfaction
- Overall calls answered/abandoned
- Calls by time

```
#=====
#      import library
#=====
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt import seaborn as sns

#=====
#      import data
#=====
data=pd.read_excel(r"C:\Users\Elbostan\Desktop\Call-Ce nter.xlsx")

#=====
#      Data preprocessing.
#=====
data.info()
data.isna().sum()
"fil
l null columns"
data[['Satisfaction rating', 'AvgTalkDuration',
'Speed of answer in seconds']] = data[['Satisfaction rating',
'AvgTalkDuration', 'Speed of answer in
seconds']].fillna(0)

describe=data.describe() plt.figure(figsize=(10, 6))
sns.heatmap(describe,annot=True, cmap="coolwarm", linewidths=0.5)
plt.show()
```



#=====

Data Analysis

#=====

" Overall customer satisfaction ? "

```
satisfaction_percentage = data['Satisfaction rating'].value_counts(normalize=True) * 100
```

```
plt.pie(satisfaction_percentage.values, labels=satisfaction_percentage.index,
autopct='%1.1f%%', startangle=90)
```

```
plt.title("satisfaction percentage")
```

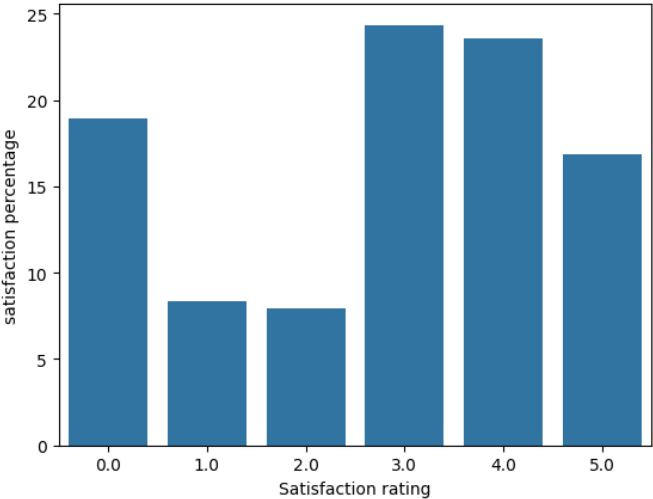
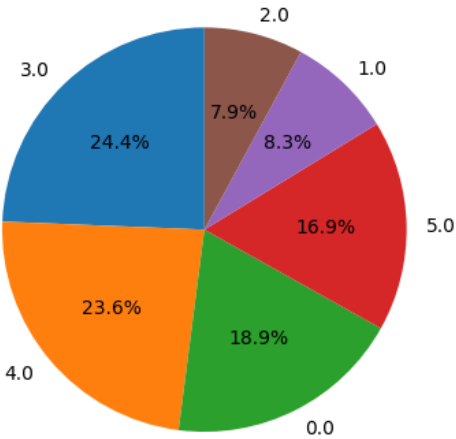
```
plt.show()
```

```
sns.barplot(x=satisfaction_percentage.index, y=satisfaction_percentage.values)
```

```
plt.ylabel('satisfaction percentage')
```

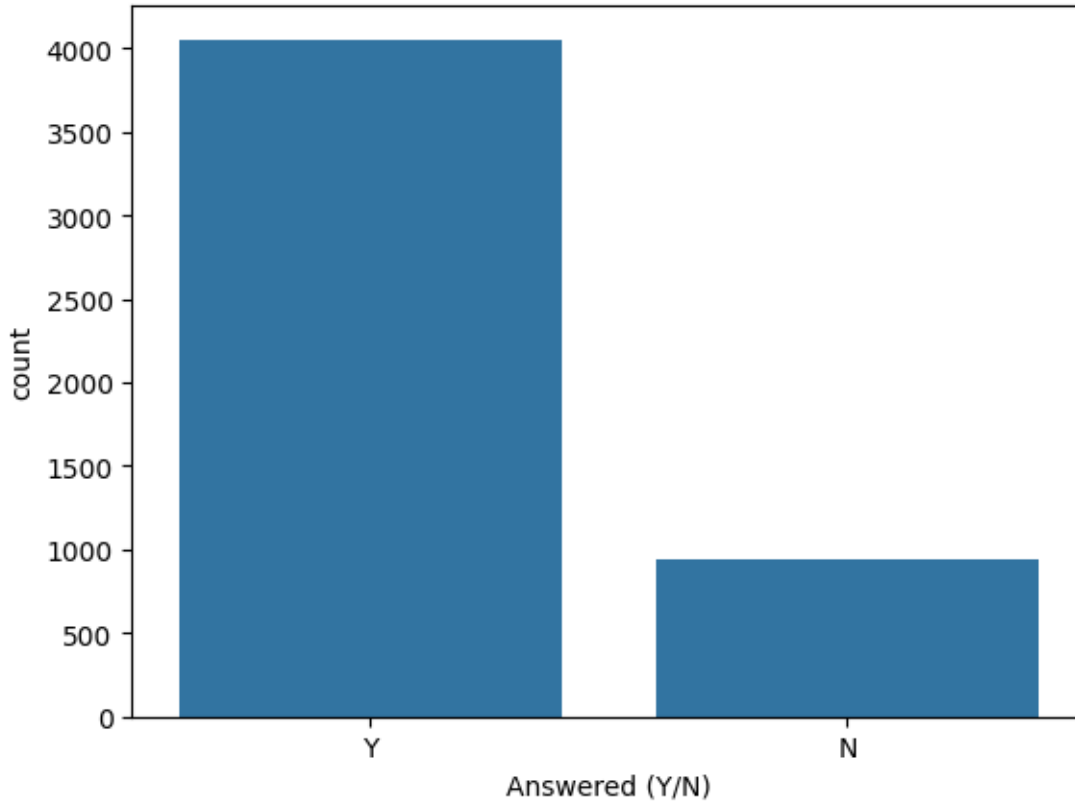
```
plt.show()
```

satisfaction percentage



Overall calls answered/abandoned ?

```
sns.countplot(data=data, x='Answered (Y/N)')
plt.show()
```



Calls by time

```
data['Date'] = pd.to_datetime(data['Date'])
data['DateTime'] = data.apply(lambda row:
pd.to_datetime(str(row['Date'].date()) + ' ' +
row['Time'].strftime('%H:%M:%S')), axis=1)
data['Hour'] = data['DateTime'].dt.hour
calls_by_hour = data.groupby('Hour')['Call Id'].count().reset_index()
calls_by_hour.rename(columns={'Call Id': 'Number of Calls'}, inplace=True)
```

```
sns.barplot(data=calls_by_hour, x='Hour', y='Number of Calls')
plt.title("calles by hours")
plt.xlabel("hours")
plt.ylabel("calles answers")
plt.xticks(rotation=45)
plt.show()
```

calles by hours

